

AI SDLC Studio

Low-Level Project Understanding & Developer Onboarding Document

Generated from the provided project codebase — September 2026

1. Project Overview

AI SDLC Studio is an AI-enabled software-development lifecycle platform. The supplied project is organized around AI/agent orchestration and controlled SDLC workflows, with human review/approval used as a progression gate where implemented.

- Goal: help structure and automate SDLC activities using AI agents while retaining human oversight.
- The supplied project context identifies requirements and design workflows as core areas.
- This document is intended for a developer joining the project without prior knowledge.
- Implementation-specific details are based on the supplied archive; unsupported details are marked as not identifiable.

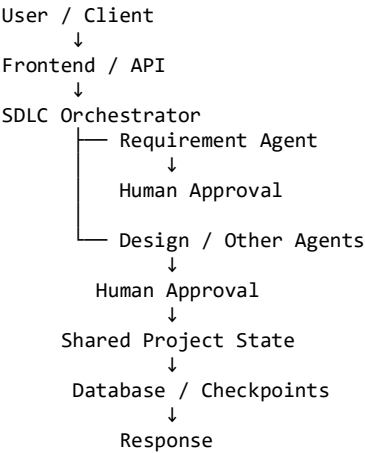
2. Technology Stack

Technology / Concept	Status	Representative files
FastAPI	Detected	AI-SDLC-Studio-version-4/README.md; AI-SDLC-Studio-version-4/backend/requirements.txt; AI-SDLC-Studio-version-4/backend/app/main.py; AI-SDLC-Studio-version-4/backend/app/agents/design_agent.py
LangGraph	Detected	AI-SDLC-Studio-version-4/README.md; AI-SDLC-Studio-version-4/backend/requirements.txt; AI-SDLC-Studio-version-4/backend/app/database.py; AI-SDLC-Studio-version-4/backend/app/agents/design_agent.py
LangChain	Detected	AI-SDLC-Studio-version-4/README.md; AI-SDLC-Studio-version-4/backend/requirements.txt; AI-SDLC-Studio-version-4/backend/app/services/project_service.py; AI-SDLC-Studio-version-4/backend/app/services/llm_provider.py
React	Detected	AI-SDLC-Studio-version-4/README.md; AI-SDLC-Studio-version-4/frontend/vite.config.ts; AI-SDLC-Studio-version-4/frontend/package.json; AI-SDLC-Studio-version-4/frontend/tsconfig.json
TypeScript	Detected	AI-SDLC-Studio-version-4/README.md; AI-SDLC-Studio-version-4/frontend/package.json; AI-SDLC-Studio-version-4/frontend/package-lock.json; AI-SDLC-Studio-version-4/backend/app/agents/design_agent.py
Tailwind	Detected	AI-SDLC-Studio-version-4/README.md; AI-SDLC-Studio-version-4/frontend/package.json; AI-SDLC-Studio-version-4/frontend/postcss.config.js; AI-SDLC-Studio-version-4/frontend/package-lock.json
SQLite	Detected	AI-SDLC-Studio-version-4/README.md; AI-SDLC-Studio-version-4/backend/alembic.ini; AI-SDLC-Studio-version-4/backend/requirements.txt; AI-SDLC-Studio-version-4/backend/app/database.py
PostgreSQL	Detected	AI-SDLC-Studio-version-4/README.md; AI-SDLC-Studio-version-4/backend/app/agents/design_agent.py; AI-SDLC-Studio-version-4/backend/app/services/project_service.py; AI-SDLC-Studio-version-4/backend/app/services/llm_provider.py
OpenAI	Detected	AI-SDLC-Studio-version-4/backend/requirements.txt
Gemini	Detected	AI-SDLC-Studio-version-4/README.md; AI-SDLC-Studio-version-4/frontend/src/App.tsx; AI-SDLC-Studio-version-4/backend/app/agents/development_agent.py; AI-SDLC-Studio-version-4/backend/app/agents/development_orchestrator.py
Pydantic	Detected	AI-SDLC-Studio-version-4/backend/requirements.txt; AI-SDLC-Studio-version-4/backend/app/schemas.py; AI-SDLC-Studio-version-4/backend/app/agents/design_agent.py; AI-SDLC-Studio-version-4/backend/app/agents/requirement_agent.py
Alembic	Detected	AI-SDLC-Studio-version-4/backend/alembic.ini; AI-SDLC-Studio-version-4/backend/alembic/env.py; AI-SDLC-Studio-version-4/backend/alembic/versions/218783eda80c_add_project_document_tables.py; AI-SDLC-Studio-version-4/backend/alembic/versions/0c02d1e20f60_add_artifact_paths_and_execution_logs.py
Requirement Agent	Detected	AI-SDLC-Studio-version-4/README.md; AI-SDLC-Studio-version-4/frontend/src/App.tsx; AI-SDLC-Studio-version-4/backend/app/agents/requirement_agent.py
Design Agent	Detected	AI-SDLC-Studio-version-4/README.md; AI-SDLC-Studio-version-

		4/frontend/src/App.tsx; AI-SDLC-Studio-version-4/backend/app/main.py; AI-SDLC-Studio-version-4/backend/app/services/agent_service.py
Development Agent	Detected	AI-SDLC-Studio-version-4/README.md; AI-SDLC-Studio-version-4/frontend/src/App.tsx; AI-SDLC-Studio-version-4/backend/app/main.py

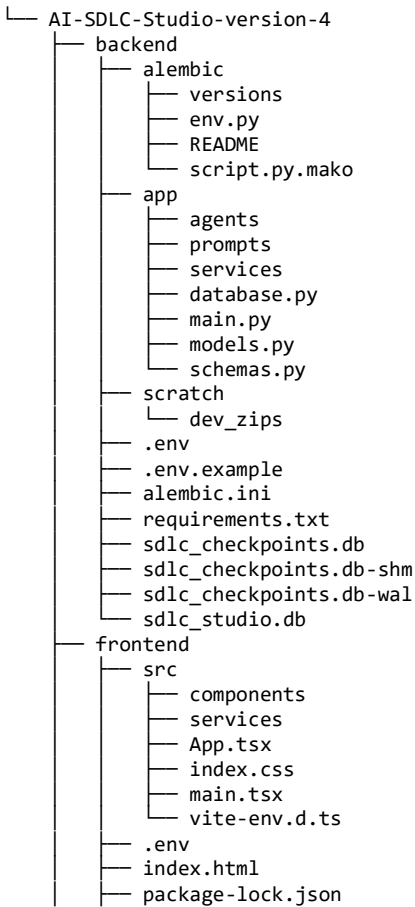
3. Architecture

The architecture is centered on an SDLC orchestration layer coordinating specialized agents and persisted workflow/project state. The supplied project context describes a Human-in-the-Loop waterfall-style progression.



Exact production deployment topology should be confirmed from the repository's deployment/configuration files.

4. Folder and File Structure



```

├── package.json
├── postcss.config.js
├── tailwind.config.js
├── tsconfig.json
├── vite.config.ts
├── .gitignore
└── README.md

```

Important files identified by filename patterns:

- AI-SDLC-Studio-version-4/README.md
- AI-SDLC-Studio-version-4/backend/sdlc_studio.db
- AI-SDLC-Studio-version-4/backend/sdlc_checkpoints.db-shm
- AI-SDLC-Studio-version-4/backend/sdlc_checkpoints.db
- AI-SDLC-Studio-version-4/backend/alembic.ini
- AI-SDLC-Studio-version-4/backend/requirements.txt
- AI-SDLC-Studio-version-4/backend/sdlc_checkpoints.db-wal
- AI-SDLC-Studio-version-4/frontend/index.html
- AI-SDLC-Studio-version-4/frontend/.env
- AI-SDLC-Studio-version-4/frontend/vite.config.ts
- AI-SDLC-Studio-version-4/frontend/package.json
- AI-SDLC-Studio-version-4/frontend/tsconfig.json
- AI-SDLC-Studio-version-4/frontend/postcss.config.js
- AI-SDLC-Studio-version-4/frontend/package-lock.json
- AI-SDLC-Studio-version-4/frontend/tailwind.config.js
- AI-SDLC-Studio-version-4/frontend/src/index.css
- AI-SDLC-Studio-version-4/frontend/src/vite-env.d.ts
- AI-SDLC-Studio-version-4/frontend/src/App.tsx
- AI-SDLC-Studio-version-4/frontend/src/main.tsx
- AI-SDLC-Studio-version-4/frontend/src/services/api.ts
- AI-SDLC-Studio-version-4/frontend/src/components/DevelopmentPanel.tsx
- AI-SDLC-Studio-version-4/backend/app/schemas.py
- AI-SDLC-Studio-version-4/backend/app/models.py
- AI-SDLC-Studio-version-4/backend/app/database.py
- AI-SDLC-Studio-version-4/backend/app/main.py
- AI-SDLC-Studio-version-4/backend/alembic/script.py.mako
- AI-SDLC-Studio-version-4/backend/alembic/env.py
- AI-SDLC-Studio-version-4/backend/alembic/README
- AI-SDLC-Studio-version-4/backend/alembic/versions/218783eda80c_add_project_document_tables.py
- AI-SDLC-Studio-version-4/backend/alembic/versions/0c02d1e20f60_add_artifact_paths_and_execution_logs.py
- AI-SDLC-Studio-version-4/backend/alembic/versions/a30555480a96_initial_revision.py
- AI-SDLC-Studio-version-4/backend/alembic/versions/a7e8bf905704_add_development_tables.py
- AI-SDLC-Studio-version-4/backend/alembic/versions/c3751fc92e5b_add_multi_agent_development_tables.py
- AI-SDLC-Studio-version-4/backend/alembic/versions/d481f9a01e23_add_testing_agent_tables.py
- AI-SDLC-Studio-version-4/backend/app/agents/state.py
- AI-SDLC-Studio-version-4/backend/app/agents/design_state.py
- AI-SDLC-Studio-version-4/backend/app/agents/design_agent.py
- AI-SDLC-Studio-version-4/backend/app/agents/development_state.py
- AI-SDLC-Studio-version-4/backend/app/agents/requirement_agent.py
- AI-SDLC-Studio-version-4/backend/app/agents/development_agent.py
- AI-SDLC-Studio-version-4/backend/app/services/document_parser.py
- AI-SDLC-Studio-version-4/backend/app/services/requirement_validator.py
- AI-SDLC-Studio-version-4/backend/app/services/project_service.py

- AI-SDLC-Studio-version-4/backend/app/services/design_service.py
- AI-SDLC-Studio-version-4/backend/app/services/input_guardrails.py
- AI-SDLC-Studio-version-4/backend/app/services/llm_provider.py
- AI-SDLC-Studio-version-4/backend/app/services/development_service.py
- AI-SDLC-Studio-version-4/backend/app/services/agent_service.py
- AI-SDLC-Studio-version-4/backend/app/services/document_generator.py
- AI-SDLC-Studio-version-4/backend/app/services/json_repair.py
- AI-SDLC-Studio-version-4/backend/app/agents/development/planning_engine.py
- AI-SDLC-Studio-version-4/backend/app/agents/development/__init__.py
- AI-SDLC-Studio-version-4/backend/app/agents/development/orchestrator.py
- AI-SDLC-Studio-version-4/backend/app/agents/development/context_builder.py
- AI-SDLC-Studio-version-4/backend/app/agents/development/development_state.py
- AI-SDLC-Studio-version-4/backend/app/agents/development/agents/__init__.py
- AI-SDLC-Studio-version-4/backend/app/agents/development/agents/self_review_agent.py
- AI-SDLC-Studio-version-4/backend/app/agents/development/agents/api_integration_agent.py
- AI-SDLC-Studio-version-4/backend/app/agents/development/agents/documentation_agent.py
- AI-SDLC-Studio-version-4/backend/app/agents/development/agents/database_agent.py
- AI-SDLC-Studio-version-4/backend/app/agents/development/agents/backend_agent.py
- AI-SDLC-Studio-version-4/backend/app/agents/development/agents/frontend_agent.py
- AI-SDLC-Studio-version-4/backend/app/agents/development/core/rules_engine.py
- AI-SDLC-Studio-version-4/backend/app/agents/development/core/merge_validation.py
- AI-SDLC-Studio-version-4/backend/app/agents/development/core/__init__.py
- AI-SDLC-Studio-version-4/backend/app/agents/development/core/code_snippet_store.py
- AI-SDLC-Studio-version-4/backend/app/agents/development/core/prompt_manager.py
- AI-SDLC-Studio-version-4/backend/app/agents/development/core/artifact_generator.py
- AI-SDLC-Studio-version-4/backend/app/agents/development/core/template_engine.py
- AI-SDLC-Studio-version-4/backend/app/agents/development/core/memory_manager.py
- AI-SDLC-Studio-version-4/backend/scratch/dev_zips/cd00eb94-592d-48e1-b684-0ba050347c01_v1/DatabaseSchema.sql
- AI-SDLC-Studio-version-4/backend/scratch/dev_zips/cd00eb94-592d-48e1-b684-0ba050347c01_v1/README.md
- AI-SDLC-Studio-version-4/backend/scratch/dev_zips/cd00eb94-592d-48e1-b684-0ba050347c01_v1/API.json

5. Backend Detailed Explanation

Method	Endpoint	Source file
GET	/	AI-SDLC-Studio-version-4/backend/app/main.py
POST	/api/projects	AI-SDLC-Studio-version-4/backend/app/main.py
GET	/api/projects	AI-SDLC-Studio-version-4/backend/app/main.py
GET	/api/projects/{project_id}	AI-SDLC-Studio-version-4/backend/app/main.py
POST	/api/projects/{project_id}/chat	AI-SDLC-Studio-version-4/backend/app/main.py
POST	/api/projects/{project_id}/approve	AI-SDLC-Studio-version-4/backend/app/main.py
GET	/api/projects/{project_id}/requirements/state	AI-SDLC-Studio-version-4/backend/app/main.py
POST	/api/projects/{project_id}/requirements/approve	AI-SDLC-Studio-version-4/backend/app/main.py
POST	/api/projects/{project_id}/requirements/reject	AI-SDLC-Studio-version-4/backend/app/main.py
POST	/api/projects/{project_id}/requirements/upload	AI-SDLC-Studio-version-4/backend/app/main.py
GET	/api/projects/{project_id}/design/state	AI-SDLC-Studio-version-4/backend/app/main.py
POST	/api/projects/{project_id}/design/approve	AI-SDLC-Studio-version-4/backend/app/main.py
POST	/api/projects/{project_id}/design/reject	AI-SDLC-Studio-version-4/backend/app/main.py
GET	/api/projects/{project_id}/status	AI-SDLC-Studio-version-4/backend/app/main.py
GET	/api/projects/{project_id}/history	AI-SDLC-Studio-version-4/backend/app/main.py
POST	/api/projects/{project_id}/design/generate	AI-SDLC-Studio-version-4/backend/app/main.py

POST	/api/projects/{project_id}/design/approve	AI-SDLC-Studio-version-4/backend/app/main.py
GET	/api/projects/{project_id}/design	AI-SDLC-Studio-version-4/backend/app/main.py
GET	/api/projects/{project_id}/design/versions	AI-SDLC-Studio-version-4/backend/app/main.py
GET	/api/projects/{project_id}/requirements/download/pdf	AI-SDLC-Studio-version-4/backend/app/main.py
GET	/api/projects/{project_id}/requirements/download/docx	AI-SDLC-Studio-version-4/backend/app/main.py
GET	/api/projects/{project_id}/requirements/download/markdown	AI-SDLC-Studio-version-4/backend/app/main.py
GET	/api/projects/{project_id}/requirements/download/json	AI-SDLC-Studio-version-4/backend/app/main.py
GET	/api/projects/{project_id}/design/download/pdf	AI-SDLC-Studio-version-4/backend/app/main.py
GET	/api/projects/{project_id}/design/download/docx	AI-SDLC-Studio-version-4/backend/app/main.py
GET	/api/projects/{project_id}/design/download/markdown	AI-SDLC-Studio-version-4/backend/app/main.py
GET	/api/projects/{project_id}/design/download/json	AI-SDLC-Studio-version-4/backend/app/main.py
POST	/api/projects/{project_id}/design/review	AI-SDLC-Studio-version-4/backend/app/main.py
POST	/api/projects/{project_id}/development/start	AI-SDLC-Studio-version-4/backend/app/main.py
POST	/api/projects/{project_id}/development/approve	AI-SDLC-Studio-version-4/backend/app/main.py
GET	/api	AI-SDLC-Studio-version-4/backend/app/services/llm_provider.py
POST	/api/checkout	AI-SDLC-Studio-version-4/backend/app/services/llm_provider.py
GET	/path	AI-SDLC-Studio-version-4/backend/app/agents/development/agents/api_integration_agent.py
POST	/path	AI-SDLC-Studio-version-4/backend/app/agents/development/agents/api_integration_agent.py
GET	/	AI-SDLC-Studio-version-4/backend/app/agents/development/core/template_engine.py

Controllers, Services, Models and Schemas

- Trace each API entry point into its service/orchestrator and then into persistence or external integrations.
- Use Pydantic/schema definitions as the authoritative request/response contract where present.
- Use actual call sites rather than filenames alone to determine business responsibilities.

6. Frontend Detailed Explanation

- AI-SDLC-Studio-version-4/frontend/index.html
- AI-SDLC-Studio-version-4/frontend/.env
- AI-SDLC-Studio-version-4/frontend/vite.config.ts
- AI-SDLC-Studio-version-4/frontend/package.json
- AI-SDLC-Studio-version-4/frontend/tsconfig.json
- AI-SDLC-Studio-version-4/frontend/postcss.config.js
- AI-SDLC-Studio-version-4/frontend/package-lock.json
- AI-SDLC-Studio-version-4/frontend/tailwind.config.js
- AI-SDLC-Studio-version-4/frontend/src/index.css
- AI-SDLC-Studio-version-4/frontend/src/vite-env.d.ts
- AI-SDLC-Studio-version-4/frontend/src/App.tsx
- AI-SDLC-Studio-version-4/frontend/src/main.tsx
- AI-SDLC-Studio-version-4/frontend/src/services/api.ts
- AI-SDLC-Studio-version-4/frontend/src/components/DevelopmentPanel.tsx

- AI-SDLC-Studio-version-4/backend/app/schemas.py
- AI-SDLC-Studio-version-4/backend/app/models.py
- AI-SDLC-Studio-version-4/backend/app/database.py
- AI-SDLC-Studio-version-4/backend/app/main.py
- AI-SDLC-Studio-version-4/backend/app/agents/state.py
- AI-SDLC-Studio-version-4/backend/app/agents/design_state.py
- AI-SDLC-Studio-version-4/backend/app/agents/design_agent.py
- AI-SDLC-Studio-version-4/backend/app/agents/development_state.py
- AI-SDLC-Studio-version-4/backend/app/agents/requirement_agent.py
- AI-SDLC-Studio-version-4/backend/app/agents/development_agent.py
- AI-SDLC-Studio-version-4/backend/app/prompts/design_prompts.py
- AI-SDLC-Studio-version-4/backend/app/prompts/requirement_prompts.py
- AI-SDLC-Studio-version-4/backend/app/services/document_parser.py
- AI-SDLC-Studio-version-4/backend/app/services/requirement_validator.py
- AI-SDLC-Studio-version-4/backend/app/services/project_service.py
- AI-SDLC-Studio-version-4/backend/app/services/design_service.py
- AI-SDLC-Studio-version-4/backend/app/services/input_guardrails.py
- AI-SDLC-Studio-version-4/backend/app/services/llm_provider.py
- AI-SDLC-Studio-version-4/backend/app/services/development_service.py
- AI-SDLC-Studio-version-4/backend/app/services/agent_service.py
- AI-SDLC-Studio-version-4/backend/app/services/document_generator.py
- AI-SDLC-Studio-version-4/backend/app/services/json_repair.py
- AI-SDLC-Studio-version-4/backend/app/agents/development/planning_engine.py
- AI-SDLC-Studio-version-4/backend/app/agents/development/__init__.py
- AI-SDLC-Studio-version-4/backend/app/agents/development/orchestrator.py
- AI-SDLC-Studio-version-4/backend/app/agents/development/context_builder.py
- AI-SDLC-Studio-version-4/backend/app/agents/development/development_state.py
- AI-SDLC-Studio-version-4/backend/app/agents/development/agents/__init__.py
- AI-SDLC-Studio-version-4/backend/app/agents/development/agents/self_review_agent.py
- AI-SDLC-Studio-version-4/backend/app/agents/development/agents/api_integration_agent.py
- AI-SDLC-Studio-version-4/backend/app/agents/development/agents/documentation_agent.py
- AI-SDLC-Studio-version-4/backend/app/agents/development/agents/database_agent.py
- AI-SDLC-Studio-version-4/backend/app/agents/development/agents/backend_agent.py
- AI-SDLC-Studio-version-4/backend/app/agents/development/agents/frontend_agent.py
- AI-SDLC-Studio-version-4/backend/app/agents/development/core/rules_engine.py
- AI-SDLC-Studio-version-4/backend/app/agents/development/core/merge_validation.py
- AI-SDLC-Studio-version-4/backend/app/agents/development/core/__init__.py
- AI-SDLC-Studio-version-4/backend/app/agents/development/core/code_snippet_store.py
- AI-SDLC-Studio-version-4/backend/app/agents/development/core/prompt_manager.py
- AI-SDLC-Studio-version-4/backend/app/agents/development/core/artifact_generator.py
- AI-SDLC-Studio-version-4/backend/app/agents/development/core/template_engine.py
- AI-SDLC-Studio-version-4/backend/app/agents/development/core/memory_manager.py

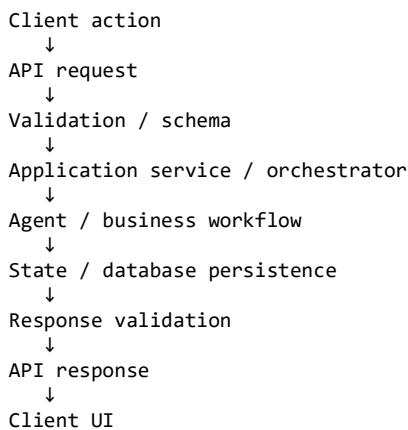
7. Database Understanding

- AI-SDLC-Studio-version-4/backend/sdlc_studio.db
- AI-SDLC-Studio-version-4/backend/sdlc_checkpoints.db-shm
- AI-SDLC-Studio-version-4/backend/sdlc_checkpoints.db
- AI-SDLC-Studio-version-4/backend/alembic.ini
- AI-SDLC-Studio-version-4/backend/sdlc_checkpoints.db-wal

- AI-SDLC-Studio-version-4/backend/app/database.py
- AI-SDLC-Studio-version-4/backend/alembic/script.py.mako
- AI-SDLC-Studio-version-4/backend/alembic/env.py
- AI-SDLC-Studio-version-4/backend/alembic/README
- AI-SDLC-Studio-version-4/backend/alembic/versions/218783eda80c_add_project_document_tables.py
- AI-SDLC-Studio-version-4/backend/alembic/versions/0c02d1e20f60_add_artifact_paths_and_execution_logs.py
- AI-SDLC-Studio-version-4/backend/alembic/versions/a30555480a96_initial_revision.py
- AI-SDLC-Studio-version-4/backend/alembic/versions/a7e8bf905704_add_development_tables.py
- AI-SDLC-Studio-version-4/backend/alembic/versions/c3751fc92e5b_add_multi_agent_development_tables.py
- AI-SDLC-Studio-version-4/backend/alembic/versions/d481f9a01e23_add_testing_agent_tables.py
- AI-SDLC-Studio-version-4/backend/app/agents/development/agents/database_agent.py
- AI-SDLC-Studio-version-4/backend/scratch/dev_zips/cd00eb94-592d-48e1-b684-0ba050347c01_v1/DatabaseSchema.sql

The supplied project context identifies SQLite as a default database with PostgreSQL support and Alembic migrations/checkpoint persistence. Exact current schema must be verified in the repository.

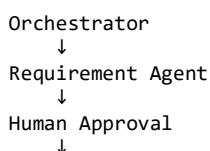
8. API and Data Flow



9. AI / LLM / Agent Architecture

- LangGraph
- LangChain
- OpenAI
- Gemini
- Pydantic
- Requirement Agent
- Design Agent
- Development Agent

The supplied project context describes a Requirement Agent producing structured requirements/SRS, a human approval gate, and a Design Agent consuming approved requirements to produce design artifacts. The implementation should remain the source of truth.



```
Design Agent
↓
Human Approval
↓
Shared Project State / Persistence
↓
Validation / Conversation State
↓
Prompt Generation
↓
LLM API
↓
Response Validation
```

10. Configuration and Environment Variables

- AI-SDLC-Studio-version-4/backend/.env
- AI-SDLC-Studio-version-4/backend/requirements.txt
- AI-SDLC-Studio-version-4/backend/.env.example
- AI-SDLC-Studio-version-4/frontend/.env
- AI-SDLC-Studio-version-4/frontend/vite.config.ts
- AI-SDLC-Studio-version-4/frontend/tsconfig.json
- AI-SDLC-Studio-version-4/frontend/postcss.config.js
- AI-SDLC-Studio-version-4/frontend/tailwind.config.js

Secrets, passwords, API keys and tokens are intentionally not reproduced.

11. Authentication and Authorization

Authentication/authorization details are not fully established by the automatic scan. If protected routes or an identity provider exist, trace middleware/configuration together with frontend token/session handling.

12. Error Handling

- AI-SDLC-Studio-version-4/backend/alembic/versions/0c02d1e20f60_add_artifact_paths_and_execution_logs.py
- AI-SDLC-Studio-version-4/backend/app/services/requirement_validator.py
- AI-SDLC-Studio-version-4/backend/app/agents/development/core/merge_validation.py
- AI-SDLC-Studio-version-4/backend/scratch/dev_zips/cd00eb94-592d-48e1-b684-0ba050347c01_v1/ChangeLog.json
- AI-SDLC-Studio-version-4/backend/scratch/dev_zips/cd00eb94-592d-48e1-b684-0ba050347c01_v1/ValidationReport.json

Confirmed error behavior should be taken from exception handlers, validators and response construction in the source.

13. Important Functions and Classes

The most important symbols should be reviewed in orchestrators, agent modules, API entry points, schemas/models, persistence modules and UI entry points.

- AI-SDLC-Studio-version-4/README.md
- AI-SDLC-Studio-version-4/backend/sdlc_studio.db
- AI-SDLC-Studio-version-4/backend/sdlc_checkpoints.db-shm
- AI-SDLC-Studio-version-4/backend/sdlc_checkpoints.db
- AI-SDLC-Studio-version-4/backend/alembic.ini
- AI-SDLC-Studio-version-4/backend/requirements.txt
- AI-SDLC-Studio-version-4/backend/sdlc_checkpoints.db-wal

- AI-SDLC-Studio-version-4/frontend/index.html
- AI-SDLC-Studio-version-4/frontend/.env
- AI-SDLC-Studio-version-4/frontend/vite.config.ts
- AI-SDLC-Studio-version-4/frontend/package.json
- AI-SDLC-Studio-version-4/frontend/tsconfig.json
- AI-SDLC-Studio-version-4/frontend/postcss.config.js
- AI-SDLC-Studio-version-4/frontend/package-lock.json
- AI-SDLC-Studio-version-4/frontend/tailwind.config.js
- AI-SDLC-Studio-version-4/frontend/src/index.css
- AI-SDLC-Studio-version-4/frontend/src/vite-env.d.ts
- AI-SDLC-Studio-version-4/frontend/src/App.tsx
- AI-SDLC-Studio-version-4/frontend/src/main.tsx
- AI-SDLC-Studio-version-4/frontend/src/services/api.ts
- AI-SDLC-Studio-version-4/frontend/src/components/DevelopmentPanel.tsx
- AI-SDLC-Studio-version-4/backend/app/schemas.py
- AI-SDLC-Studio-version-4/backend/app/models.py
- AI-SDLC-Studio-version-4/backend/app/database.py
- AI-SDLC-Studio-version-4/backend/app/main.py
- AI-SDLC-Studio-version-4/backend/alembic/script.py.mako
- AI-SDLC-Studio-version-4/backend/alembic/env.py
- AI-SDLC-Studio-version-4/backend/alembic/README
- AI-SDLC-Studio-version-4/backend/alembic/versions/218783eda80c_add_project_document_tables.py
- AI-SDLC-Studio-version-4/backend/alembic/versions/0c02d1e20f60_add_artifact_paths_and_execution_logs.py
- AI-SDLC-Studio-version-4/backend/alembic/versions/a30555480a96_initial_revision.py
- AI-SDLC-Studio-version-4/backend/alembic/versions/a7e8bf905704_add_development_tables.py
- AI-SDLC-Studio-version-4/backend/alembic/versions/c3751fc92e5b_add_multi_agent_development_tables.py
- AI-SDLC-Studio-version-4/backend/alembic/versions/d481f9a01e23_add_testing_agent_tables.py
- AI-SDLC-Studio-version-4/backend/app/agents/state.py
- AI-SDLC-Studio-version-4/backend/app/agents/design_state.py
- AI-SDLC-Studio-version-4/backend/app/agents/design_agent.py
- AI-SDLC-Studio-version-4/backend/app/agents/development_state.py
- AI-SDLC-Studio-version-4/backend/app/agents/requirement_agent.py
- AI-SDLC-Studio-version-4/backend/app/agents/development_agent.py

14. Complete Feature-Wise Flow

Feature	High-level flow
Requirement workflow	Input → Requirement Agent → structured requirements/SRS → human approval → persisted state.
Design workflow	Approved requirements → Design Agent → design/SDD artifacts → human approval → persisted state.
Validation workflow	Input/output validation around workflow and LLM interactions where implemented.
Project state	Project metadata, requirement versions, human reviews and activity/state information are persisted according to the implemented data model.

15. Dependencies

- AI-SDLC-Studio-version-4/backend/requirements.txt
- AI-SDLC-Studio-version-4/frontend/package.json

- AI-SDLC-Studio-version-4/frontend/package-lock.json

AI-SDLC-Studio-version-4/backend/requirements.txt

```
fastapi>=0.100.0
uvicorn>=0.22.0
sqlalchemy>=2.0.0
pydantic>=2.0.0
langchain>=0.1.0
langchain-community>=0.0.10
langchain-openai>=0.0.2
langchain-google-genai>=0.0.6
langgraph>=0.0.10
langgraph-checkpoint-sqlite>=2.0.0
python-dotenv>=1.0.0
reportlab>=4.0.0
python-docx>=1.2.0
pypdf>=4.0.0
python-multipart>=0.0.9
```

AI-SDLC-Studio-version-4/frontend/package.json

```
{
  "name": "ai-sdlc-studio",
  "private": true,
  "version": "1.0.0",
  "type": "module",
  "scripts": {
    "dev": "vite",
    "build": "tsc && vite build",
    "preview": "vite preview"
  },
  "dependencies": {
    "react": "^18.2.0",
    "react-dom": "^18.2.0",
    "lucide-react": "^0.294.0",
    "mermaid": "^10.9.0"
  },
  "devDependencies": {
    "@types/react": "^18.2.37",
    "@types/react-dom": "^18.2.15",
    "@vitejs/plugin-react": "^4.2.0",
    "typescript": "^5.2.2",
    "vite": "^5.0.0",
    "tailwindcss": "^3.3.5",
    "postcss": "^8.4.31",
    "autoprefixer": "^10.4.16"
  }
}
```

AI-SDLC-Studio-version-4/frontend/package-lock.json

```
{
  "name": "ai-sdlc-studio",
  "version": "1.0.0",
  "lockfileVersion": 3,
  "requires": true,
  "packages": {
    "": {
      "name": "ai-sdlc-studio",
      "version": "1.0.0",
      "dependencies": {
        "lucide-react": "^0.294.0",
        "mermaid": "^10.9.0",
        "react": "^18.2.0",
        "react-dom": "^18.2.0"
      },
      "devDependencies": {
        "@types/react": "^18.2.37",
        "@types/react-dom": "^18.2.15",
        "@vitejs/plugin-react": "^4.2.0",
        "autoprefixer": "^10.4.16",
        "postcss": "^8.4.31",
        "tailwindcss": "^3.3.5",
        "typescript": "^5.2.2",
        "vite": "^5.0.0"
      }
    }
  },
}
```

```

"node_modules/@alloc/quick-lru": {
  "version": "5.2.0",
  "resolved": "https://registry.npmjs.org/@alloc/quick-lru/-/quick-lru-5.2.0.tgz",
  "integrity": "sha512-UrcABB+4bUrFABwbluTIBErXwvbsU/V7TZWfmbgJfbkwiBuziS9gxd0DUyuiecfDGQ85jg1MW6juS3+z5TsKLw==",
  "dev": true,
  "license": "MIT",
  "engines": {
    "node": ">=10"
  },
  "funding": {
    "url": "https://github.com/sponsors/sindresorhus"
  }
},
"node_modules/@babel/code-frame": {
  "version": "7.29.7",
  "resolved": "https://registry.npmjs.org/@babel/code-frame/-/code-frame-7.29.7.tgz",
  "integrity": "sha512-BsIMbbsGwX3DZkVWZgXtQh3HjYpdQ9xjO2cFWq+iPvZrj1t7X0mStvJFuzrT8xMzuEP56arW/31iZAnZSA==",
  "dev": true,
  "license": "MIT",
  "dependencies": {
    "@babel/helper-validator-identifier": "^7.29.7",
    "js-tokens": "^4.0.0",
    "picocolors": "^1.1.1"
  },
  "engines": {
    "node": ">=6.9.0"
  }
},
"node_modules/@babel/compat-data": {
  "version": "7.29.7",
  "resolved": "https://registry.npmjs.org/@babel/compat-data/-/compat-data-7.29.7.tgz",
  "integrity": "sha512-fp8e03H9G4nHIAoIeGUi5xO+j8UGbB8wSs60+D1wHEL56o8XUYr2p10CjHh0G8p5oYjW84tUwYV1MAA==",
  "dev": true,
  "license": "MIT",
  "engines": {
    "node": ">=6.9.0"
  }
},
"node_modules/@babel/core": {
  "version": "7.29.7",
  "resolved": "https://registry.npmjs.org/@babel/core/-/core-7.29.7.tgz",
  "integrity": "sha512-7eYTWDiGnC/3hpeUnz9rK6Bp2U7+RfZxwR2clQE0uXn4PbLPSKl1I05Ys88ZtkVw0Ck0ZKX28HxhYXU8vA==",
  "dev": true,
  "license": "MIT",
  "dependencies": {
    "@babel/code-frame": "^7.29.7",
    "@babel/generator": "^7.29.7",
    "@babel/helper-compilation-targets": "^7.29.7",
    "@babel/helper-module-transforms": "^7.29.7",
    "@babel/helpers": "^7.29.7",
    "@babel/parser": "^7.29.7",
    "@babel/template": "^7.29.7",
    "@babel/traverse": "^7.29.7",
    "@babel/types": "^7.29.7",
  }
}

```

16. Setup and Installation

- Read the repository README and dependency manifests first.
- Create environment configuration using the project's templates/examples; never copy real secrets into documentation.
- Install backend/frontend dependencies with the package managers specified by the repository.
- Initialize/migrate the database if migration scripts exist.
- Start required backend/orchestration services, then the frontend/client.
- Verify a representative SDLC workflow end-to-end.

Exact commands are intentionally not invented when they cannot be established from the supplied repository.

17. Running the Project

- AI-SDLC-Studio-version-4/README.md
- AI-SDLC-Studio-version-4/frontend/package.json
- AI-SDLC-Studio-version-4/backend/app/main.py
- AI-SDLC-Studio-version-4/backend/alembic/README
- AI-SDLC-Studio-version-4/backend/scratch/dev_zips/cd00eb94-592d-48e1-b684-0ba050347c01_v1/README.md

18. Testing

- AI-SDLC-Studio-version-4/backend/alembic/versions/d481f9a01e23_add_testing_agent_tables.py

Important test areas: API contracts, workflow transitions, approval/rejection paths, persistence, validation failures and LLM failure/retry behavior.

19. Deployment / Cloud

- No deployment-specific files identified by the filename scan.

20. Security

- Keep API keys, passwords and tokens outside source control.
- Validate API inputs and structured agent outputs.
- Protect human-approval endpoints and project-state transitions.
- Use least-privilege access for external AI/database services.
- Treat generated LLM output as untrusted until validated.
- Separate implemented controls from recommended security improvements.

21. Important Design Decisions

- Human-in-the-Loop gating is a central workflow concept in the supplied project context.
- SDLC work is structured into specialized workflow stages/agents rather than one monolithic AI call.
- Structured schemas/state make agent outputs and transitions explicit.
- Database/checkpoint persistence preserves workflow/project state.
- Where a design rationale is not documented, the reason is not identifiable from the codebase.

22. Known Issues and Technical Debt

- Verify environment variables and safe configuration examples.
- Add/verify tests around every workflow transition and approval gate.
- Check for duplicate prompt/orchestration logic.
- Ensure malformed LLM output cannot corrupt persisted state.
- Review logs for accidental exposure of prompts, secrets or sensitive data.
- Review migration and API contract test coverage.

23. Developer Onboarding Guide

If I Join This Project Today

- Start with README and repository tree.

- Understand project state/data model.
- Trace the main orchestrator/workflow graph.
- Read Requirement Agent and schemas/prompts.
- Read human-review/approval implementation.
- Read Design Agent and its input/output contract.
- Trace persistence/checkpoint behavior.
- Trace API endpoints into services/workflows.
- Run a small end-to-end workflow.
- Only then modify individual modules.

30-minute path

- 0–5 min: README + tree.
- 5–10 min: entry point + configuration.
- 10–18 min: orchestrator + project state.
- 18–24 min: Requirement/Design flow.
- 24–30 min: run one representative workflow.

2-hour deep dive

- 30–60 min: agent/workflow modules, schemas and validation.
- 60–85 min: API, database/checkpoints and migrations.
- 85–105 min: frontend/client flow and error states.
- 105–120 min: tests, deployment and known gaps.

24. Glossary

Term	Meaning
SDLC	Software Development Life Cycle.
Human-in-the-Loop	A workflow where a human reviews/approves an AI-generated artifact before progression.
Requirement Agent	Agent responsible for transforming requirement inputs into structured requirement/SRS artifacts where implemented.
Design Agent	Agent consuming approved requirements to produce design artifacts where implemented.
Orchestrator	Component coordinating workflow stages/agents and state transitions.
SRS	Software Requirements Specification.
SDD	Software Design Document.
LLM	Large Language Model.
Checkpoint	Persisted workflow state used to resume or inspect execution.

25. Final Project Summary

Project in One Paragraph

AI SDLC Studio is an AI-assisted SDLC platform designed to structure software engineering work into explicit workflow stages and specialized agents. Its key architectural idea is controlled progression through requirements and design with human review, structured state, validation and persistence.

Project in 10 Steps

1. Create/select a project and provide SDLC input.
2. Validate and normalize incoming data.

3. Start the appropriate SDLC workflow/orchestrator.
4. Generate structured requirements through the Requirement Agent.
5. Persist/version the requirement artifact.
6. Pause for human review/approval.
7. Pass approved requirements to the Design Agent.
8. Generate and validate design artifacts.
9. Pause for the next human approval/state transition where required.
10. Persist final workflow state and expose results to the client.

Key Files to Remember

- AI-SDLC-Studio-version-4/README.md
- AI-SDLC-Studio-version-4/backend/sdlc_studio.db
- AI-SDLC-Studio-version-4/backend/sdlc_checkpoints.db-shm
- AI-SDLC-Studio-version-4/backend/sdlc_checkpoints.db
- AI-SDLC-Studio-version-4/backend/alembic.ini
- AI-SDLC-Studio-version-4/backend/requirements.txt
- AI-SDLC-Studio-version-4/backend/sdlc_checkpoints.db-wal
- AI-SDLC-Studio-version-4/frontend/index.html
- AI-SDLC-Studio-version-4/frontend/.env
- AI-SDLC-Studio-version-4/frontend/vite.config.ts
- AI-SDLC-Studio-version-4/frontend/package.json
- AI-SDLC-Studio-version-4/frontend/tsconfig.json
- AI-SDLC-Studio-version-4/frontend/postcss.config.js
- AI-SDLC-Studio-version-4/frontend/package-lock.json
- AI-SDLC-Studio-version-4/frontend/tailwind.config.js
- AI-SDLC-Studio-version-4/frontend/src/index.css
- AI-SDLC-Studio-version-4/frontend/src/vite-env.d.ts
- AI-SDLC-Studio-version-4/frontend/src/App.tsx
- AI-SDLC-Studio-version-4/frontend/src/main.tsx
- AI-SDLC-Studio-version-4/frontend/src/services/api.ts
- AI-SDLC-Studio-version-4/frontend/src/components/DevelopmentPanel.tsx
- AI-SDLC-Studio-version-4/backend/app/schemas.py
- AI-SDLC-Studio-version-4/backend/app/models.py
- AI-SDLC-Studio-version-4/backend/app/database.py
- AI-SDLC-Studio-version-4/backend/app/main.py
- AI-SDLC-Studio-version-4/backend/alembic/script.py.mako
- AI-SDLC-Studio-version-4/backend/alembic/env.py
- AI-SDLC-Studio-version-4/backend/alembic/README
- AI-SDLC-Studio-version-4/backend/alembic/versions/218783eda80c_add_project_document_tables.py
- AI-SDLC-Studio-version-4/backend/alembic/versions/0c02d1e20f60_add_artifact_paths_and_execution_logs.py

Key APIs to Remember

Metho d	Endpoint	File
GET	/	AI-SDLC-Studio-version-4/backend/app/main.py
POST	/api/projects	AI-SDLC-Studio-version-4/backend/app/main.py
GET	/api/projects	AI-SDLC-Studio-version-4/backend/app/main.py
GET	/api/projects/{project_id}	AI-SDLC-Studio-version-4/backend/app/main.py
POST	/api/projects/{project_id}/chat	AI-SDLC-Studio-version-4/backend/app/main.py
POST	/api/projects/{project_id}/approve	AI-SDLC-Studio-version-4/backend/app/main.py
GET	/api/projects/{project_id}/requirements/state	AI-SDLC-Studio-version-4/backend/app/main.py

POST	/api/projects/{project_id}/requirements/approve	AI-SDLC-Studio-version-4/backend/app/main.py
POST	/api/projects/{project_id}/requirements/reject	AI-SDLC-Studio-version-4/backend/app/main.py
POST	/api/projects/{project_id}/requirements/upload	AI-SDLC-Studio-version-4/backend/app/main.py
GET	/api/projects/{project_id}/design/state	AI-SDLC-Studio-version-4/backend/app/main.py
POST	/api/projects/{project_id}/design/approve	AI-SDLC-Studio-version-4/backend/app/main.py
POST	/api/projects/{project_id}/design/reject	AI-SDLC-Studio-version-4/backend/app/main.py
GET	/api/projects/{project_id}/status	AI-SDLC-Studio-version-4/backend/app/main.py
GET	/api/projects/{project_id}/history	AI-SDLC-Studio-version-4/backend/app/main.py
POST	/api/projects/{project_id}/design/generate	AI-SDLC-Studio-version-4/backend/app/main.py
POST	/api/projects/{project_id}/design/approve	AI-SDLC-Studio-version-4/backend/app/main.py
GET	/api/projects/{project_id}/design	AI-SDLC-Studio-version-4/backend/app/main.py
GET	/api/projects/{project_id}/design/versions	AI-SDLC-Studio-version-4/backend/app/main.py
GET	/api/projects/{project_id}/requirements/download/pdf	AI-SDLC-Studio-version-4/backend/app/main.py
GET	/api/projects/{project_id}/requirements/download/docx	AI-SDLC-Studio-version-4/backend/app/main.py
GET	/api/projects/{project_id}/requirements/download/markdown	AI-SDLC-Studio-version-4/backend/app/main.py
GET	/api/projects/{project_id}/requirements/download/json	AI-SDLC-Studio-version-4/backend/app/main.py
GET	/api/projects/{project_id}/design/download/pdf	AI-SDLC-Studio-version-4/backend/app/main.py
GET	/api/projects/{project_id}/design/download/docx	AI-SDLC-Studio-version-4/backend/app/main.py
GET	/api/projects/{project_id}/design/download/markdown	AI-SDLC-Studio-version-4/backend/app/main.py
GET	/api/projects/{project_id}/design/download/json	AI-SDLC-Studio-version-4/backend/app/main.py
POST	/api/projects/{project_id}/design/review	AI-SDLC-Studio-version-4/backend/app/main.py
POST	/api/projects/{project_id}/development/start	AI-SDLC-Studio-version-4/backend/app/main.py
POST	/api/projects/{project_id}/development/approve	AI-SDLC-Studio-version-4/backend/app/main.py
GET	/api	AI-SDLC-Studio-version-4/backend/app/services/llm_provider.py
POST	/api/checkout	AI-SDLC-Studio-version-4/backend/app/services/llm_provider.py
GET	/path	AI-SDLC-Studio-version-4/backend/app/agents/development/agents/api_integration_agent.py
POST	/path	AI-SDLC-Studio-version-4/backend/app/agents/development/agents/api_integration_agent.py
GET	/	AI-SDLC-Studio-version-4/backend/app/agents/development/core/template_engine.py

Key Database Tables

- The supplied project context identifies projects, requirements, requirement_versions, human_reviews and activity_logs as important persisted concepts.
- Verify exact current definitions in schema/migration files.

Key Technologies

- FastAPI
- LangGraph
- LangChain
- React
- TypeScript
- Tailwind
- SQLite
- PostgreSQL

- OpenAI
- Gemini
- Pydantic
- Alembic
- Requirement Agent
- Design Agent
- Development Agent

26. Source of Truth

The repository source code, dependency manifests, migrations, configuration templates and deployment definitions are the source of truth. This document is an onboarding aid and should be regenerated after major architecture changes.

- AI-SDLC-Studio-version-4/README.md
- AI-SDLC-Studio-version-4/backend/sdlc_studio.db
- AI-SDLC-Studio-version-4/backend/sdlc_checkpoints.db-shm
- AI-SDLC-Studio-version-4/backend/sdlc_checkpoints.db
- AI-SDLC-Studio-version-4/backend/alembic.ini
- AI-SDLC-Studio-version-4/backend/requirements.txt
- AI-SDLC-Studio-version-4/backend/sdlc_checkpoints.db-wal
- AI-SDLC-Studio-version-4/frontend/index.html
- AI-SDLC-Studio-version-4/frontend/.env
- AI-SDLC-Studio-version-4/frontend/vite.config.ts
- AI-SDLC-Studio-version-4/frontend/package.json
- AI-SDLC-Studio-version-4/frontend/tsconfig.json
- AI-SDLC-Studio-version-4/frontend/postcss.config.js
- AI-SDLC-Studio-version-4/frontend/package-lock.json
- AI-SDLC-Studio-version-4/frontend/tailwind.config.js
- AI-SDLC-Studio-version-4/frontend/src/index.css
- AI-SDLC-Studio-version-4/frontend/src/vite-env.d.ts
- AI-SDLC-Studio-version-4/frontend/src/App.tsx
- AI-SDLC-Studio-version-4/frontend/src/main.tsx
- AI-SDLC-Studio-version-4/frontend/src/services/api.ts
- AI-SDLC-Studio-version-4/frontend/src/components/DevelopmentPanel.tsx
- AI-SDLC-Studio-version-4/backend/app/schemas.py
- AI-SDLC-Studio-version-4/backend/app/models.py
- AI-SDLC-Studio-version-4/backend/app/database.py
- AI-SDLC-Studio-version-4/backend/app/main.py
- AI-SDLC-Studio-version-4/backend/alembic/script.py.mako
- AI-SDLC-Studio-version-4/backend/alembic/env.py
- AI-SDLC-Studio-version-4/backend/alembic/README
- AI-SDLC-Studio-version-4/backend/alembic/versions/218783eda80c_add_project_document_tables.py
- AI-SDLC-Studio-version-4/backend/alembic/versions/0c02d1e20f60_add_artifact_paths_and_execution_logs.py
- AI-SDLC-Studio-version-4/backend/alembic/versions/a30555480a96_initial_revision.py
- AI-SDLC-Studio-version-4/backend/alembic/versions/a7e8bf905704_add_development_tables.py
- AI-SDLC-Studio-version-4/backend/alembic/versions/c3751fc92e5b_add_multi_agent_development_tables.py
- AI-SDLC-Studio-version-4/backend/alembic/versions/d481f9a01e23_add_testing_agent_tables.py
- AI-SDLC-Studio-version-4/backend/app/agents/state.py
- AI-SDLC-Studio-version-4/backend/app/agents/design_state.py
- AI-SDLC-Studio-version-4/backend/app/agents/design_agent.py

- AI-SDLC-Studio-version-4/backend/app/agents/development_state.py
- AI-SDLC-Studio-version-4/backend/app/agents/requirement_agent.py
- AI-SDLC-Studio-version-4/backend/app/agents/development_agent.py
- AI-SDLC-Studio-version-4/backend/app/services/document_parser.py
- AI-SDLC-Studio-version-4/backend/app/services/requirement_validator.py
- AI-SDLC-Studio-version-4/backend/app/services/project_service.py
- AI-SDLC-Studio-version-4/backend/app/services/design_service.py
- AI-SDLC-Studio-version-4/backend/app/services/input_guardrails.py
- AI-SDLC-Studio-version-4/backend/app/services/llm_provider.py
- AI-SDLC-Studio-version-4/backend/app/services/development_service.py
- AI-SDLC-Studio-version-4/backend/app/services/agent_service.py
- AI-SDLC-Studio-version-4/backend/app/services/document_generator.py
- AI-SDLC-Studio-version-4/backend/app/services/json_repair.py

27. Document Generation Notes

Generated from the supplied AI-SDLC-Studio-version-4.zip archive and the supplied low-level documentation requirements. Sensitive credentials are omitted, and unsupported assumptions are not presented as implemented functionality.